Sentinel — <TEAM NAME>

સેન્ટિનલ — ગુજરાત પોલીસ માટે · Gujarat Police Innovation Challenge 2026 · Grand finale, i-Hub Gandhinagar, 12–13 October 2026

What Sentinel is. One registry, one map and one viewer for Gujarat's government cameras, with plate reading where a camera's view allows it and vehicle and person counts everywhere else. It is Model 2 (unified viewing and selective analytics) on the mandatory Model 1 foundation (registry and GIS): it connects to cameras, recorders and VMS re-streams directly, installs nothing on a department's side and replaces nothing. Video stays with the departments; only metadata (plate reads, counts, alerts) crosses the network, and every search and operator action is audited. No facial recognition, by design.

The live test, step by step

Step	What the judges see	Where in Sentinel
Onboard	A government camera onboarded live from its stream address: probe → relay → register, about 2 s; the rest as a batch	Registry → *Add camera by stream URL*; `scripts/onboard_urls.py`
View	Every camera on one map and one video wall, H.264 and H.265 alike; Health grades each camera's plate-reading quality and says what would improve it	Map · Video wall · Health & Reports
Read	Plate reads with vehicle type and colour; vehicle and person counts per camera per minute, on every camera at once	Overview · Health & Reports
Alert	A watchlist entry raises a live alert with its evidence crop about a second after the first read; near misses labelled; a read below confidence 0.9 says "check the crop"	Watchlist · Alerts
Trace	The registration given on the spot: a 24-hour look-back with one retrospective alert, then its journey as stops and legs, near misses marked, implausible legs flagged	Vehicle search → journey view
Report	Detected plates with timestamps as PDF and CSV (a SHA-256 per crop); the session's audit trail	`scripts/detection_report.py` · Audit

Four numbers, with their conditions

Number	Condition
30 of 30 government cameras onboarded, 1.0–2.2 s each	The organisers' sandbox, 25 Sep 2026, by RTSP address; H.264 and H.265, 960×576 to 2560×1440; all 30 viewed and counted, 4 at a time in rotation on one laptop (`bench/results/sandbox.md`)
90 of 120 real Gujarat plates stored exactly right (75.0%); 107 within one character	Blind check by one team member on the Plate review page, 26 Sep, on the 2 of 30 views where plates are readable (CAM06, CAM07). Precision on stored reads only: capture on real plates is unmeasured. At Sentinel's confidence ≥ 0.9, 69 of 81 exact (`sandbox_review.md`). On stock footage with synthetic plates: 96.8% exact, 91.2% capture (`anpr_eval_current.md`)
1.06 s median from the first read of a vehicle to its alert (p95 1.94 s)	31 alerts, 4 cameras, worker and API on one laptop and one clock, 24 Sep; excludes camera transport and the browser (`latency.md`)
40 streams counted at once on one laptop at 2.46 CPU cores, GPU 13% busy	Counts-only worker from keyframes, 0.78 frame/s per stream, 7 Oct; no plate worker running at the same time; 50 streams saturate; count accuracy unmeasured (`counts_capacity.md`)

Architecture

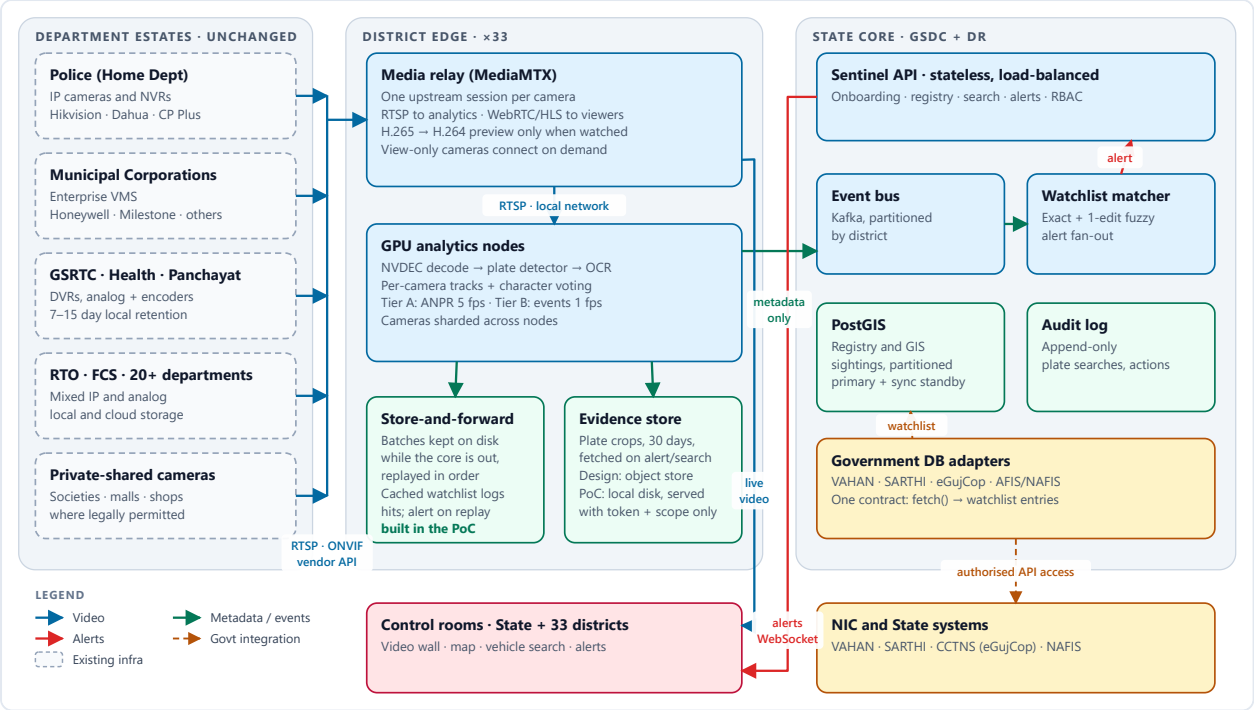


Figure 1. Department estates untouched; district edges relay and analyse; only metadata reaches the state core.

Sized for 80,000 cameras by a capacity model (`bench/capacity.py` , Plan for Scale): 248 edge GPU nodes, about 2.7 Mbps of metadata statewide against 320 Gbps to carry every stream centrally, ₹100.7 crore over five years (₹210 per camera per month). Tier shares, streams per node and unit prices are assumptions, labelled as such; the pilot measures them. The core was run at statewide size on one laptop: 706 sightings a second matched against 100,000 watchlist entries, a floor, not a sizing (`core_scale.md`).

Said plainly

- **VAHAN and eGujCop are mock adapters** over fixture files behind one adapter contract; SARTHI, AFIS and NAFIS are declared, awaiting access.
- **Plates are read on 2 of the 30 sandbox views.** The other 28 are overview or red-light views where plates are too small; they are counted, and the Health page says what would make them readable (zoom, shutter, infrared, re-aim). The grade describes the camera; it does not predict whether a given read is right.
- **People are counted, never identified.** An intrusion alert holds the zone, the count and the dwell, never a box, image or track.
- **Since the submission (26 Sep → 10 Oct):** the counts-only worker and the two-laptop set-up were built and measured; the "check the crop" confidence marking passed its pre-registered gate on 8 Oct and is on; the OCR fine-tune was non-inferior on real plates and not adopted; the YOLOX detectors were not trained (the pilot's plan). Every accuracy, capture and latency figure above still describes the build that runs.

Contacts

<TEAM NAME> · <name, role, phone, email> · <name, role, phone, email> · Documents: HLD, Plan for Scale, privacy note, demo videos, `bench/results/` (every figure's conditions)